

The angle between a face and the base

HANDOUT



Interactive version

HOOK

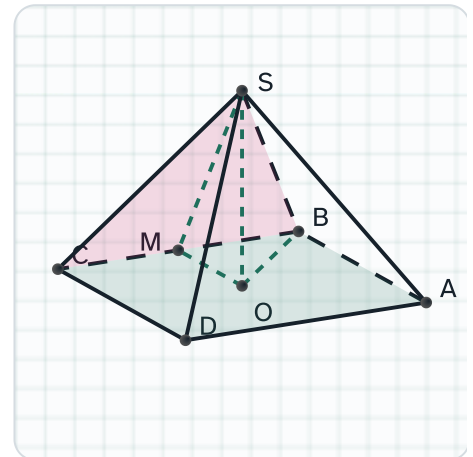
Which angle?

SABCD is a regular pyramid: its base is a square, and the apex S is directly above the centre O of the base. M is the midpoint of BC. Turn the figure if it helps.

Level 2 • satisfactory

Which angle is the angle between the side face SBC and the base?

- ☐ A $\angle BSC$
☐ B $\angle SMO$
☐ C $\angle SBO$
☐ D $\angle SBC$



Face SBC and the base are shaded.

EXPLORE

Taller pyramid, steeper face?

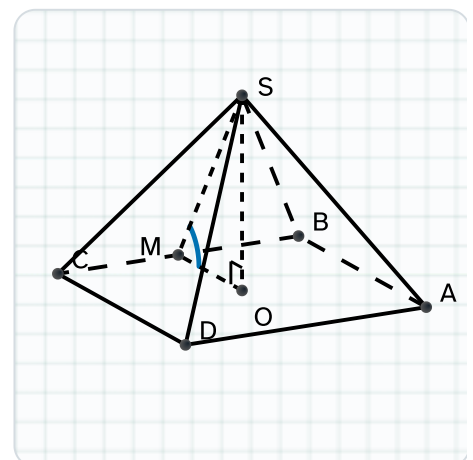
Level 2 • satisfactory

Predict first. The base stays the same and the pyramid gets taller. What happens to the angle φ between a side face and the base? Then drag the height slider to check.

- ☐ A it stays the same
 ☐ B it shrinks
☐ C it grows

Level 3 • basic

The base side is 4. At which height is $\varphi = 45^\circ$? Try it with the slider, then enter the exact value.



$\varphi = \angle SMO \approx 56.3^\circ$ (height 3, base side 4).

EXPLAIN

Finding the angle, step by step

The task

SABCD is a regular pyramid. Its base is a square with side $a = 4$ and its height is $SO = 3$.

Find the angle between the side face **SBC** and the **base**. Both planes are shaded.

Find the common edge

The two planes meet along the edge **BC**.

The angle between two planes is measured at one point of their common edge, between two lines that are both perpendicular to that edge, one in each plane.

Perpendicular in the base

Take **M**, the midpoint of BC. O is the centre of the square, so **OM** is parallel to AB and $OM \perp BC$.

$$OM = \frac{a}{2} = 2.$$

Perpendicular in the side face

Triangle SBC is isosceles ($SB = SC$), so its median **SM** is also its height: $SM \perp BC$.

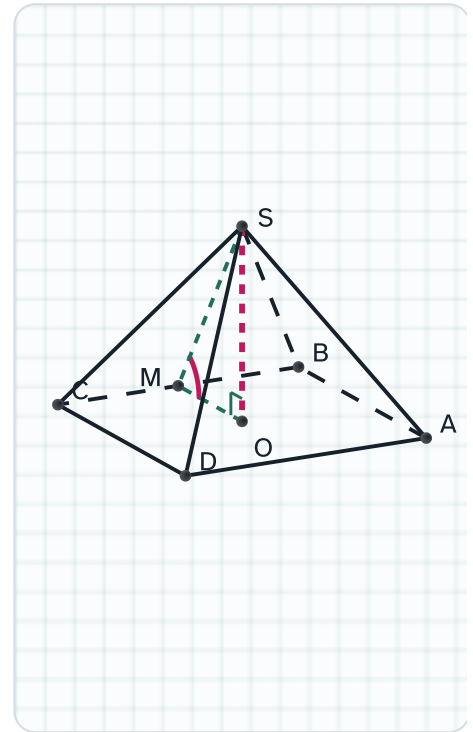
Now both lines are perpendicular to BC at the same point M.

The angle

So $\varphi = \angle SMO$ is the angle between the planes. Triangle SOM has a right angle at O:

$$\tan \varphi = \frac{SO}{OM} = \frac{3}{2}, \quad \varphi \approx 56.3^\circ.$$

Press **Look along BC** under the figure: when B and C coincide, you see the angle at its true size.



Three worked examples

Example 1 • Given the apothem

TASK

A regular quadrilateral pyramid has base side 6 and apothem (height of a side face) $SM = 5$. Find the angle between a side face and the base.

THE ANGLE

As in the explanation, the angle is $\varphi = \angle SMO$, with $OM = \frac{6}{2} = 3$.

TRIANGLE SOM

$$\cos \varphi = \frac{OM}{SM} = \frac{3}{5}, \text{ so } \varphi \approx 53.1^\circ.$$

Example 2 • Given the angle, find the height

TASK

A regular quadrilateral pyramid has base side 10. The angle between a side face and the base is 60° . Find the height.

THE ANGLE

$\varphi = \angle SMO = 60^\circ$ and $OM = 5$.

TRIANGLE SOM

$$SO = OM \cdot \tan 60^\circ = 5\sqrt{3} \approx 8.66.$$

Example 3 • Given the lateral edge

TASK

A regular quadrilateral pyramid has base side 6 and lateral edge $SB = 5$. Find the angle between a side face and the base.

HEIGHT

OB is half the diagonal: $OB = 3\sqrt{2}$. In triangle SOB: $SO = \sqrt{25 - 18} = \sqrt{7}$.

TRIANGLE SOM

$$OM = 3, \text{ so } \tan \varphi = \frac{\sqrt{7}}{3} \text{ and } \varphi \approx 41.4^\circ.$$

SUMMARY

The method

To find the angle between two planes:

1. Find their **common edge**.
2. Pick one point on it. In a regular pyramid, take the midpoint M of a base edge.
3. Draw a **perpendicular to the edge in each plane**: OM in the base, SM in the face.
4. The angle between these two lines is the angle between the planes. Solve the right triangle SOM.

Do not confuse it with the angle between an **edge** and the base: that one is $\angle SBO$, between SB and its projection OB.